

Jay Rao

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EDUCATION

Northwestern University, Evanston, IL

Expected June 2028

Bachelor of Arts (B.A.) in Computer Science, Concentration in Artificial Intelligence

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Design & Analysis of Algorithms, Scalable Software Architecture

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, SQL

Frameworks & Libraries: FastAPI, Node.js/Express, React, PyTorch, CatBoost, NumPy, Pandas, Playwright, pytest

Tools & Platforms: Git/GitHub, GitHub Actions (CI/CD), AWS (EC2, ECS, VPC, ALB, RDS, S3, SES, IAM, Secrets Manager), Terraform, Docker, PostgreSQL, MySQL, Linux/Bash, Agile/Scrum

EXPERIENCE

Software Engineering Intern | Lastrada Music Technology, Inc.

Sep. 2026 – Present

- Building an LLM-driven intake agent for a music sample-clearance platform (Node.js, TypeScript, PostgreSQL) that parses emails and PDF letters into structured requests, matches sampled songs to the rights catalog and routes each to the correct rights holder
- Scoping turnkey third-party product integrations and a Roam-based virtual office for multi-agent workflow orchestration

Founding Engineer | GridGate LLC

Feb. 2026 – Present

- Sole engineer on GridGate Queue Analysis v1.0, an end-to-end data platform built around a 19-source data lake fed by 20+ scrapers, a 9-module feature pipeline, and a modeling layer covering 3,700+ MISO interconnection projects
- Automated ingestion of 300,000+ state and county regulatory filings and meeting minute PDFs across 15 states and 1,000+ counties using fault-tolerant scrapers with retry logic plus LLM-driven discovery for non-standard sites
- Trained and validated two CatBoost classifiers covering project success and withdrawal (0.93 ROC-AUC, 85% accuracy) and 5-class withdrawal phase (0.94 one-vs-rest ROC-AUC), each with SHAP explainability for stakeholders
- Designed a 6-tier entity resolution pipeline that maps 3,700+ interconnection projects to their parent developers at 83% precision using document-grounded lookups and multi-stage evidence verification against public filings

Technology Manager | iloveyou LLC

Feb. 2026 – Sep. 2026

- Lead developer managing a team of 3 junior engineers through sprint planning, task assignment and code review, taking a consumer social app from an empty repository to a shipped v1.0 production release across backend and iOS
- Built the backend from scratch in Node.js/Express on PostgreSQL with JWT auth and refresh token rotation, email and phone verification (Amazon SES, Twilio Verify), APNs push notifications and REST APIs
- Engineered AWS infrastructure as code in Terraform (VPC, ECS Fargate, ALB, RDS, S3, CloudFront, SES, Secrets Manager, IAM) with remote state and DynamoDB locking, plus CI/CD pipelines in GitHub Actions

PROJECTS

game.sense | Python, PyTorch, PyTorch Geometric, FastAPI, PyQt6, pytest

- Building a spatial intelligence engine that predicts ball and vehicle positions up to 5 seconds ahead from a single occluded viewpoint, chaining a 4-layer GATv2 graph encoder into per-entity LSTMs and a Gaussian mixture decoder trained on 2,193 parsed replays
- Cut 1 second ball position error from an 11.39m baseline to 3.06m over five retraining rounds, sharing one frustum and occlusion masking core between offline label generation and live inference so training and serving geometry can never silently drift apart
- Shipped the live path as a Kalman filter for visible entities plus frame-budgeted model inference behind a PyQt6 overlay and a FastAPI WebSocket server, validated by a 277-test pytest suite spanning separate CUDA and CPU training/inference environments

nuPython Interpreter & Debugger | C, C++, STL, Valgrind

- Built an interpreter for a Python-like language in 2,000+ lines of C/C++ with a recursive expression evaluator, int/float coercion, control flow, built-in functions and runtime type checking with descriptive semantic errors
- Designed a simulated RAM pairing a dynamically resizing cell array with an alphabetically sorted name-to-address map, giving $O(\log n)$ binary search lookup with deep copying and leak-free teardown verified under Valgrind on every run
- Implemented a GDB-style source-level debugger that unlinks and relinks the program graph to single-step execution, with support for live variable inspection, memory dumps and full execution history tracking

vibes.fm | React, Node.js, Express, MySQL, AWS, Transformers.js, OAuth 2.0

- Engineered a full-stack playlist generator that embeds track metadata, audio features and lyrics into a 384-dimensional vector, then ranks an in-memory cache by cosine similarity and exports matches to Spotify via PKCE OAuth 2.0, deployed on EC2 behind nginx